

Kitaev Honeycomb model

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The Kitaev honeycomb model is one exactly solvable 2 dimensional interacting spin model which is rare in 2 dimensional systems. It has direction dependent bond strengths with spin interactions. The key point to exactly solving it is the conserved plaquette vortices and static $\mathbb{Z}_2$ gauge field obtained by four Majorana representation. The Hamiltonian is read as a quadratic form by such point to be exactly solvable. The ground state is realized by Lieb's theorem on vortex-free sector, the phase diagrams contains three gapped phases and one gapless phase with two Dirac cones. A weak magnetic field is applied by third order of perturbation to open the gap and make it topological. It is then realized as a three-spin term. By four Majorana representation, it becomes a next nearest neighbour Majorana hopping which breaks time reversal symmetry, opens the gap and yields non-zero Chern number.

I. INTRODUCTION

In two dimension, it is rare to find exactly solvable models. The Kitaev honeycomb model [1] is one exactly solvable model, initially as a platform for non-Abelian anyons with rich physics much beyond the original goal: its ground state is a $\mathbb{Z}_2$ quantum spin liquid, its excitations are Majorana fermions and static $\mathbb{Z}_2$ gauge fluxes. In a magnetic field a chiral topological phase is realized on the gapped phase.

The key to exact solving it is the conserved $\mathbb{Z}_2$ plaquette fluxes: written in terms of four Majorana operators per spin, they become a static gauge field and the model reduces to a graphene-like free-fermion problem. After presenting the solution (Sections II to IV), we compute the phase diagram, the Chern number $\mathcal{C} = \pm 1$ (similar with the Haldane model [2] but different type of particles). All details about calculation and further explanations can be found in Supplemental Material [3].

II. THE MODEL

The Hamiltonian of the model is (Fig. 1a) [1]:

$$\hat{H} = -J_x \sum_{\langle j,k \rangle_x} \sigma_j^x \sigma_k^x - J_y \sum_{\langle j,k \rangle_y} \sigma_j^y \sigma_k^y - J_z \sum_{\langle j,k \rangle_z} \sigma_j^z \sigma_k^z \quad (1)$$

with bond-direction dependent interaction coefficients J_x, J_y, J_z . The operator $\hat{K}_{jk} = \sigma_j^\alpha \sigma_k^\alpha$, $\alpha \in \{x, y, z\}$ is defined as bond operator, with α determined by the direction of the bond. The plaquette operator is defined as:

$$\hat{W}_p = \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z = \hat{K}_{12} \hat{K}_{23} \hat{K}_{34} \hat{K}_{45} \hat{K}_{56} \hat{K}_{61}. \quad (2)$$

p is the label of the plaquette. The $\hat{W}_p$ commute with one another and with every bond operator $\hat{K}_{jk}$ (any two overlapping operators share an even number of anticommuting Pauli matrices), hence $[\hat{W}_p, \hat{H}] = 0$, i.e., the $\hat{W}_p$

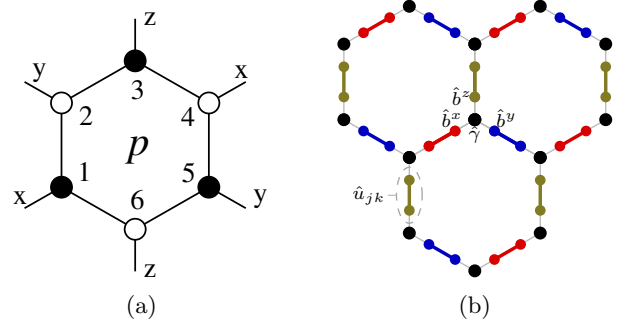


FIG. 1: (a) A plaquette p : The bond strengths depend on x, y, z directions, filled site is sublattice, blank site is even sublattice; the site numbering $1, \dots, 6$ and the outgoing bond labels are those entering the flux operator $\hat{W}_p$ of Eq. (2). (b) Four-Majorana representation: each spin becomes four Majoranas $\hat{\gamma}, \hat{b}^x, \hat{b}^y, \hat{b}^z$ (x red, y blue, z olive); on every α -bond the Majorana bond operator: $\hat{u}_{jk} = i \hat{b}_j^\alpha \hat{b}_k^\alpha$.

are conserved, and since $\hat{W}_p^2 = \hat{1}$ the Hilbert space splits into sectors labelled by $w_p = \pm 1$ on every plaquette. Each plaquette has six vertices and each vertex is shared by three plaquettes, so n vertices give $n/2$ plaquettes: the 2^n -dimensional Hilbert space splits into $2^{n/2}$ flux sectors of dimension $2^{n/2}$ each. This still can not fix the problem, the so-called parton construction gives the key to exact solution.

III. REPRESENTING SPINS BY MAJORANA OPERATORS

Kitaev represents each spin by four Majorana operators $\hat{b}^x, \hat{b}^y, \hat{b}^z, \hat{\gamma}$, shown in Fig. 1b. Majorana operators as a decomposition of complex fermionic operator: $\hat{f} = \hat{c} + \hat{c}^\dagger$, $\hat{f}^\dagger = (\hat{c} - \hat{c}^\dagger)/i$. This enlarges the dimension of space on each site: 2 dimensional Fock space of one spin to 4 dimensional Fock space of four Majoranas.

Pauli operators becomes $\tilde{\sigma}^\alpha = i\hat{b}^\alpha\hat{\gamma}$ and The bond operators become $\hat{K}_{jk} = -i\hat{u}_{jk}\hat{\gamma}_j\hat{\gamma}_k$ with $\hat{u}_{jk} := i\hat{b}_j^\alpha\hat{b}_k^\alpha$ which is Hermitian and squares to one. To fix the dimension enlarged problem, operator $\hat{D} := \hat{b}^x\hat{b}^y\hat{b}^z\hat{\gamma}$ is used to determine the physical subspace: a state $|\xi\rangle$ is physical iff $\hat{D}|\xi\rangle = |\xi\rangle$. One can find $[\hat{D}, \hat{H}] = 0$, which leaves physical subspace invariant.

IV. SPECTRUM AND PHASE DIAGRAM

With the Majorana bond operators $\hat{u}_{jk} = i\hat{b}_j^\alpha\hat{b}_k^\alpha$ defined above, the Hamiltonian becomes a quadratic form of free Majorana fermions, $\tilde{H} = \frac{i}{4} \sum_{j,k} \hat{A}_{jk} \hat{\gamma}_j \hat{\gamma}_k$, with $\hat{A}_{jk} = 2J_{\alpha_{jk}} \hat{u}_{jk}$ on connected sites and zero otherwise. $\hat{u}_{jk}$ commutes with $\tilde{H}$ and has eigenvalues ± 1 , forms a static $\mathbb{Z}_2$ gauge field. Once a configuration $\{u_{jk}\}$ is fixed, $\tilde{H}$ can be exactly diagonalized; from its ground state $|\tilde{\Psi}_u\rangle$ the physical state is recovered by the projection $|\Psi_w\rangle = [\prod_j (1 + \hat{D}_j)/2] |\tilde{\Psi}_u\rangle$. The label w is the gauge-equivalence class of $\{u_{jk}\}$, which is characterized by the fluxes $w_p = \prod_{(j,k) \in \partial p} u_{jk} = \pm 1$ (j odd, k even). There are many possible configurations of $\{u_{jk}\}$, by Lieb's theorem [4] the ground state lies in the flux-free sector with $w_p = +1$ on every plaquette, and we can choose the simplest gauge, $u_{jk} = +1$ on all links.

The Bravais lattice of honeycomb lattice is not the honeycomb itself, but half of the lattice. We place the odd (A) sublattice at the cell origin $\mathbf{R}$ and reach the even (B) sublattice by the bond vectors δ_α ($A \rightarrow B$), with $\mathbf{n}_1 = \delta_z - \delta_y$ and $\mathbf{n}_2 = \delta_z - \delta_x$; explicit conventions and geometry are given in the SM [3], Fig. S1. The real-space Hamiltonian in the $u_{jk} = 1$ gauge is

$$\tilde{H}_u = i \sum_{\mathbf{R}, \alpha} J_\alpha \hat{\gamma}_{\mathbf{R}}^A \hat{\gamma}_{\mathbf{R}+\delta_\alpha}^B. \quad (3)$$

Applying fourier transformation in the periodic gauge and choosing the sublattice spinor $\chi_{\mathbf{q}} = (\tilde{\gamma}_{\mathbf{q}}^A, \tilde{\gamma}_{\mathbf{q}}^B)^T$, the Hamiltonian in Bloch form is

$$\tilde{H}_u = \frac{1}{2} \sum_{\mathbf{q}} \chi_{\mathbf{q}}^\dagger h(\mathbf{q}) \chi_{\mathbf{q}}, \quad h(\mathbf{q}) = \begin{pmatrix} 0 & if(\mathbf{q}) \\ -if(\mathbf{q})^* & 0 \end{pmatrix}, \quad (4)$$

with factor $f(\mathbf{q}) = \sum_\alpha J_\alpha e^{i\mathbf{q} \cdot \delta_\alpha}$. Expanding in Pauli matrices, $h(\mathbf{q}) = \mathbf{d}(\mathbf{q}) \cdot \boldsymbol{\sigma}$ gives $d_x = -\text{Im } f$, $d_y = -\text{Re } f$, and $d_z = 0$. The vanishing of d_z reflects the chiral (sublattice) symmetry: $\mathbf{d}$ stays in the xy -plane over the whole Brillouin zone. The bulk dispersion is

$$\epsilon_\pm(\mathbf{q}) = \pm |f(\mathbf{q})| = \pm \sqrt{d_x^2 + d_y^2}. \quad (5)$$

The gap closes at $f(\mathbf{q}) = 0$. Geometrically, each term $J_\alpha e^{i\mathbf{q} \cdot \delta_\alpha}$ is a vector of length $|J_\alpha|$ in the complex

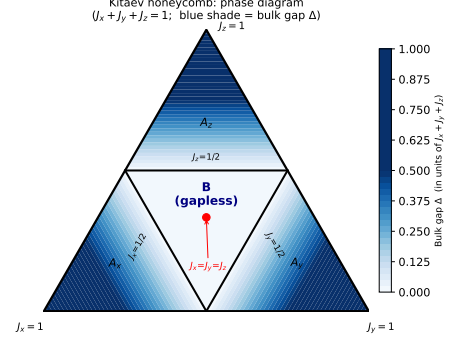


FIG. 2: Phase diagram by constraint $J_x + J_y + J_z = 1$: the central triangle is the gapless phase B, bounded by $J_\alpha = \frac{1}{2}$; the corners three are the gapped phases A_x, A_y, A_z . Shading: bulk gap Δ . In phase B it is clearly gapless.

plane; their sum vanishes only if the three vectors close into a triangle, which requires the triangle inequalities $|J_x| \leq |J_y| + |J_z|$ (and cyclic). When the inequalities hold strictly the spectrum is gapless (phase B); when one is violated, it is gapped (phase A_α); see Fig. 2.

At the symmetric point $J_x = J_y = J_z$, the three bonds become equivalent and the Hamiltonian acquires a C_3 rotational symmetry. Since the corners of the Brillouin zone are the fixed points of C_3 , the solutions of $f(\mathbf{q}) = 0$ are pinned there: the two Dirac points locate exactly at the inequivalent conical points $\pm \mathbf{K}$, and linearizing $f(\mathbf{q})$ about them yields a graphene-like linear dispersion (SM [3], Fig. S2).

Chiral symmetry constrain $d_z(\mathbf{q})$ strictly to be zero, so gap closing only requires $d_x = d_y = 0$: two conditions on two dimensional Brillouin zone, generically satisfied at the Dirac points. A nonzero d_z adds a third condition, which is impossible to find the solution, so breaking the chiral symmetry gaps the Dirac points and phase B becomes gapped.

V. TIME-REVERSAL BREAKING AND THE CHERN NUMBER

The time reversal operator acts on the Hamiltonian Eq. (4) as $\hat{T} = \sigma_z \mathcal{K}$, the detail could be find in SM [3]. The result: $\sigma_z h^*(\mathbf{q}) \sigma_z = h(-\mathbf{q})$, and matching Pauli matrices gives

$$d_x(-\mathbf{q}) = -d_x(\mathbf{q}), \quad d_y(-\mathbf{q}) = d_y(\mathbf{q}), \quad d_z(-\mathbf{q}) = +d_z(\mathbf{q}). \quad (6)$$

Thus time-reversal symmetry forces $d_z(\mathbf{q})$ to be even: a time-reversal-symmetric mass gives the two Dirac points equal masses and the Chern number vanishes (shown be-

low); a topological gap needs an odd mass, the time-reversal breaking mass, which requires broken time-reversal symmetry.

A weak external magnetic field $\mathbf{h} = (h_x, h_y, h_z)$ breaks time-reversal symmetry and enters as the perturbation $V = -\sum_j \mathbf{h} \cdot \boldsymbol{\sigma}_j$; for simplicity, we work at the isotropic point $J_x = J_y = J_z \equiv J$. Since $\sigma_j^\alpha = i\hat{b}_j^\alpha \gamma_j$ carries a single gauge Majorana, which anticommutes with the link operator $\hat{u}_{jk}$ on the α -bond at site j , meaning $\hat{u}_{jk}$ is flipped by σ ; the bond is shared by two plaquettes, so both w_p flip from $+1$ to -1 and the state leaves the vortex-free sector.

To obtain the effective Hamiltonian within the vortex-free sector we use the Schrieffer-Wolff (perturbative) expansion in V ,

$$H_{\text{eff}} = \sum_{n=1}^{\infty} H_{\text{eff}}^{(n)}, \quad H_{\text{eff}}^{(n)} = \Pi_0 V (G'_0 V)^{n-1} \Pi_0, \quad (7)$$

where $\Pi_0 = \prod_p \frac{1+\hat{W}_p}{2} \prod_j \frac{1+\hat{D}_j}{2}$ projects onto the vortex-free sector and G'_0 is the unperturbed Green's function with that sector removed; since the intermediate two-vortex states cost an energy of order $|J|$, $G'_0(E_0) \simeq -(1 - \Pi_0)/|J|$. Each factor of V breaks time reversal symmetry, so the n -th order term picks up $(-1)^n$ under $\hat{T}$: only odd orders can generate the time-reversal breaking mass d_z .

At first order, $H_{\text{eff}}^{(1)} = \Pi_0 V \Pi_0 = 0$: a single σ_j^α flips w_p on the two plaquettes adjacent to its bond and takes the state out of the vortex-free sector, so the left first Π_0 projects it to zero. At second order, $H_{\text{eff}}^{(2)} = \Pi_0 V G'_0 V \Pi_0 \neq 0$, since the second perturbation term can annihilate the vortex pair created by the first. However, with $n = 2$ factors of $\mathbf{h}$ it is still time-reversal symmetric and can not make the gap topological, so we drop it. The leading time-reversal breaking term is the third order, $H_{\text{eff}}^{(3)} = \Pi_0 V G'_0 V G'_0 V \Pi_0$. The only possible configuration is three spin interactions at nearest three bonds, or 3 bond near the same site (SM [3], Fig. S3). With two Green's functions $G'_0 \sim -1/|J|$ and three powers of $\mathbf{h}$ it takes the form

$$H_{\text{eff}}^{(3)} \sim -\kappa \sum_{\langle jkl \rangle} \sigma_j^x \sigma_k^y \sigma_l^z, \quad \kappa \sim \frac{h^x h^y h^z}{|J|^2}, \quad (8)$$

where in each triple the central spin σ_l^z sits on an even (B) site l , and σ_j^x, σ_k^y sit on the two odd (A) sites linked to l by the x - and y -bonds.

Inserting the Majorana spin expression σ_j^α and $\hat{D} = 1$ into third order term, with a fixed gauge $\{u = 1\}$, gives a next-nearest neighbour hopping between same-sublattice sites, $\sigma_j^x \sigma_k^y \sigma_l^z = -i\hat{\gamma}_j \hat{\gamma}_k$ with j, k displaced by $\mathbf{n}_1 - \mathbf{n}_2$ (geometry in the [3], Fig. S4) which is identical to the structure of the Haldane model [2]. We sum over the lattice. An A -centred process carries the opposite sign on

the B sublattice, since the directed bond reverses. So the same-sublattice hopping contributes the diagonal term $d_z(\mathbf{q}) \sigma_z$ to the Bloch Hamiltonian. Fourier transforming ([3]) gives

$$d_z(\mathbf{q}) = 6\kappa \left[\sin(\mathbf{q} \cdot \mathbf{n}_1) + \sin(\mathbf{q} \cdot \mathbf{n}_2) + \sin(\mathbf{q} \cdot (\mathbf{n}_1 - \mathbf{n}_2)) \right], \quad (9)$$

As a sum of sines, d_z is odd, $d_z(-\mathbf{q}) = -d_z(\mathbf{q})$: the field breaks time-reversal symmetry, and the Bloch Hamiltonian becomes

$$h(\mathbf{q}) = \begin{pmatrix} d_z(\mathbf{q}) & if(\mathbf{q}) \\ -if^*(\mathbf{q}) & -d_z(\mathbf{q}) \end{pmatrix} = \mathbf{d}(\mathbf{q}) \cdot \boldsymbol{\sigma}, \quad (10)$$

Gapped spectrum. For $h = \mathbf{d} \cdot \boldsymbol{\sigma}$ the eigenvalues are $\pm |\mathbf{d}|$. Since $d_x^2 + d_y^2 = |f(\mathbf{q})|^2$ as previous solution, the dispersion is

$$\varepsilon_{\pm}(\mathbf{q}) = \pm \sqrt{|f(\mathbf{q})|^2 + d_z(\mathbf{q})^2}. \quad (11)$$

As the solution given in gapless phase B, $|f(\mathbf{q})|^2$ still vanishes at the two Dirac points $\pm \mathbf{K}$. With non-zero mass term the gap at the Dirac points separates as:

$$d_z(\pm \mathbf{K}) = \mp 3\sqrt{3}\kappa, \quad \Delta = 2|d_z(\mathbf{K})| = 6\sqrt{3}|\kappa| \quad (12)$$

and the time-reversal breaking mass term will also make the gap topological.

Chern number. The topological invariant of a gapped two-dimensional system of noninteracting fermions is the Chern number of the occupied band [5],

$$\mathcal{C} = \frac{1}{2\pi} \int_{\text{BZ}} \Omega(\mathbf{q}) d^2q, \quad (13)$$

where $\Omega(\mathbf{q})$ is the Berry curvature of the negative-energy band. The integral reduces to a sum of local contributions [3],

$$\mathcal{C} = \frac{1}{2} \sum_{i=\pm \mathbf{K}} \text{sgn}(m_i) \nu_i, \quad (14)$$

where $\text{sgn}(m_{\pm \mathbf{K}}) = \mp \text{sgn}(\kappa)$ is the mass and $\nu_i = \pm 1$ the winding number at each Dirac point. Linearizing f gives $\nu_{\pm \mathbf{K}} = \mp 1$: the two windings are opposite, since the total winding over the Brillouin-zone torus must vanish [6]. The masses are opposite as well, $\text{sgn}(m_{\pm \mathbf{K}}) = \mp \text{sgn}(\kappa)$. Hence $\mathcal{C} = \text{sgn}(\kappa) = \pm 1$.

VI. CONCLUSION

The conserved plaquette vortices split the Hilbert space into flux sectors. Together with the four Majorana representation, the model becomes free Majorana fermions in a static $\mathbb{Z}_2$ gauge field. A triangular phase

diagram is realized and contains one gapless phase similar to graphene with two Dirac points. A weak magnetic field breaks the chiral and time reversal symmetry. At third order in perturbation theory it generates a next nearest neighbour hopping which gaps the two Dirac cones. The gapped phase carries a topological invariant, the Chern number $\mathcal{C} = \pm 1$.

AI usage: Claude (Anthropic) was used as an assistant in this project. For the figures, it provided coding help; the choice of what to plot and the approach are my own. It also helped with the L^AT_EX template and the formatting of the report and supplementary material. The physics, derivations, and text are my own.

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